Elementary & Intermediate Algebra

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A bag contains 8 red marbles, 6 green marbles, and 10 blue marbles. If 1 marble is chosen at random from the bag, what is the probability that the marble is NOT red?

To find the probability that the chosen marble is NOT red, determine the number of favorable outcomes and divide by the total number of outcomes. 1. Find the total number of marbles in the bag: $8 + 6 + 10 = 24$ 2. Find the number of marbles that are not red (the green and blue marbles): $6 + 10 = 16$ Alternatively, use the complement rule: $P(\text{not red}) = 1 - P(\text{red}) = 1 - \frac{8}{24} = 1 - \frac{1}{3} = \frac{2}{3}$ 3. Calculate the probability: $P(\text{not red}) = \frac{16}{24} = \frac{2}{3}$ Common Misconceptions: - $\frac{1}{3}$ is the probability of selecting a red marble ($\frac{8}{24}$), failing to account for the word NOT. - $\frac{1}{2}$ or $\frac{3}{4}$ typically result from counting only one of the non-red colors or miscounting the total number of marbles.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the standard (x, y) coordinate plane, the line segment with endpoints $A(1, -2)$ and $B(5, 6)$ is perpendicularly bisected by line ℓ. If the point $(9, k)$ lies on line ℓ, what is the value of k?

To find the value of k, determine the equation of the perpendicular bisector line ℓ in two main steps: 1. Find the midpoint of $\overline{AB}$: Line ℓ bisects segment $\overline{AB}$, so it passes through the midpoint M: $M = \left(\frac{1+5}{2}, \frac{-2+6}{2}\right) = (3, 2)$ 2. Find the slope of line ℓ: The slope of $\overline{AB}$ is: $m_{AB} = \frac{6-(-2)}{5-1} = \frac{8}{4} = 2$ Since line ℓ is perpendicular to $\overline{AB}$, its slope is the negative reciprocal of 2: $m_\ell = -\frac{1}{2}$ 3. Substitute the point $(9, k)$ into the line equation: Using point-slope form with midpoint $(3, 2)$: $y - 2 = -\frac{1}{2}(x - 3)$ Since $(9, k)$ lies on ℓ: $k - 2 = -\frac{1}{2}(9 - 3) = -\frac{1}{2}(6) = -3$ $k = 2 - 3 = -1$

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The function $f(x) = a\sin(bx) + c$ has a maximum value of 9, a minimum value of -1, and a period of $\frac{\pi}{2}$. If $a > 0$ and $b > 0$, what is the value of $a + b + c$?

To find the value of $a + b + c$, determine each parameter from the given properties of the sinusoidal function: 1. Amplitude a and Vertical Shift c: Since $-1 \le \sin(bx) \le 1$ and $a > 0$, the maximum and minimum values of $f(x)$ are: $\text{Maximum} = c + a = 9$ $\text{Minimum} = c - a = -1$ Adding these two equations gives: $2c = 8 \Rightarrow c = 4$ Substituting $c = 4$ into $c + a = 9$: $4 + a = 9 \Rightarrow a = 5$ 2. Frequency Parameter b: The period T of a sine function $\sin(bx)$ with $b > 0$ is given by $T = \frac{2\pi}{b}$. Setting this equal to the given period $\frac{\pi}{2}$: $\frac{2\pi}{b} = \frac{\pi}{2} \Rightarrow b\pi = 4\pi \Rightarrow b = 4$ 3. Sum of the Parameters: $a + b + c = 5 + 4 + 4 = 13$

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, what is the distance between the points $(-1, 2)$ and $(3, 5)$?

To find the distance between two points (x_1, y_1) and (x_2, y_2) in the standard (x, y) coordinate plane, apply the distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ Substitute the coordinates of $(-1, 2)$ and $(3, 5)$: $x_2 - x_1 = 3 - (-1) = 4$ $y_2 - y_1 = 5 - 2 = 3$ Compute the distance: $d = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5$ Common Misconceptions: - $\sqrt{7}$ comes from incorrectly subtracting rather than adding the squares: $\sqrt{4^2 - 3^2} = \sqrt{7}$. - 7 represents the sum of the horizontal and vertical displacements $(4 + 3 = 7)$, confusing straight-line distance with Manhattan distance. - 25 is the value of $d^2 = 4^2 + 3^2$, forgetting to take the square root.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle k, 3 \rangle$, where $k > 0$. If the magnitude of the vector sum $\mathbf{u} + \mathbf{v}$ is $\sqrt{29}$, what is the value of k?

To find k, follow these steps: 1. Find the vector sum $\mathbf{u} + \mathbf{v}$: Add corresponding components: $\mathbf{u} + \mathbf{v} = \langle 2 + k, -5 + 3 \rangle = \langle k + 2, -2 \rangle$ 2. Express the magnitude of $\mathbf{u} + \mathbf{v}$: The magnitude of a vector $\langle x, y \rangle$ is given by $\sqrt{x^2 + y^2}$: $||\mathbf{u} + \mathbf{v}|| = \sqrt{(k + 2)^2 + (-2)^2} = \sqrt{(k + 2)^2 + 4}$ 3. Set up and solve the equation: Given that the magnitude is $\sqrt{29}$: $\sqrt{(k + 2)^2 + 4} = \sqrt{29}$ Square both sides: $(k + 2)^2 + 4 = 29$ $(k + 2)^2 = 25$ Taking the square root of both sides gives $k + 2 = \pm 5$. Since $k > 0$, we take the positive root: $k + 2 = 5 \Rightarrow k = 3$

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given the matrices $A = [2, -3; 1, 4]$ and $B = [-5, 1; 3, -2]$, what is the matrix $2A + B$?

To find $2A + B$, first perform scalar multiplication by multiplying each entry of matrix A by 2: $2A = 2[2, -3; 1, 4] = [2(2), 2(-3); 2(1), 2(4)] = [4, -6; 2, 8]$ Next, add matrix B to $2A$ by adding corresponding entries: $2A + B = [4, -6; 2, 8] + [-5, 1; 3, -2] = [4 + (-5), -6 + 1; 2 + 3, 8 + (-2)] = [-1, -5; 5, 6]$ Common errors: - Simply computing $A + B$ gives $[-3, -2; 4, 2]$. - Computing $2A - B$ gives $[9, -7; -1, 10]$. - Forgetting to multiply one of the entries (such as the bottom-left entry) by 2 yields $[-1, -5; 4, 6]$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, what are the coordinates of the midpoint of a line segment with endpoints $(-3, 8)$ and $(5, -2)$?

To find the midpoint (x_m, y_m) of a line segment with endpoints (x_1, y_1) and (x_2, y_2), use the midpoint formula: $x_m = \frac{x_1 + x_2}{2}, \quad y_m = \frac{y_1 + y_2}{2}$ Substitute the given coordinates $(-3, 8)$ and $(5, -2)$: $x_m = \frac{-3 + 5}{2} = \frac{2}{2} = 1$ $y_m = \frac{8 + (-2)}{2} = \frac{6}{2} = 3$ Thus, the midpoint is $(1, 3)$. Common errors: - Finding the difference instead of the average yields $\left(\frac{5 - (-3)}{2}, \frac{-2 - 8}{2}\right) = (4, -5)$. - Forgetting to divide the sum by 2 yields $(2, 6)$. - Reversing the x- and y-coordinates yields $(3, 1)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If $g(x) = 2x^2 - 5x + 1$, what is the value of $g(-3)$?

To find $g(-3)$, substitute $x = -3$ into the given function expression: $g(-3) = 2(-3)^2 - 5(-3) + 1$ Follow the standard order of operations: 1. Evaluate the exponent: $(-3)^2 = 9$. 2. Perform the multiplications: $2(9) = 18$ and $-5(-3) = +15$. 3. Add the terms: $18 + 15 + 1 = 34$. Common Misconceptions: - 4 results from an error with negative signs on the linear term (or evaluating $g(3) = 18 - 15 + 1 = 4$). - -2 results from improperly squaring the negative input as $-3^2 = -9$, yielding $2(-9) + 15 + 1 = -2$. - -32 combines both sign errors: $-18 - 15 + 1 = -32$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the standard (x, y) coordinate plane, what is the midpoint of the line segment with endpoints $(3, -4)$ and $(-5, 8)$?

To find the midpoint of a line segment with endpoints (x_1, y_1) and (x_2, y_2), use the midpoint formula: $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ Substitute the coordinates of the endpoints $(3, -4)$ and $(-5, 8)$: $x_m = \frac{3 + (-5)}{2} = \frac{-2}{2} = -1$ $y_m = \frac{-4 + 8}{2} = \frac{4}{2} = 2$ Thus, the midpoint is $(-1, 2)$. Common errors: - $(-2, 4)$ results from adding coordinates without dividing by 2. - $(1, -2)$ results from sign errors during addition. - $(4, -6)$ results from subtracting coordinates rather than averaging them.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A fair coin is tossed twice. Find $P(\text{two heads})$.

Because each coin flip is an independent event, apply the multiplication rule for independent probabilities: $P(\text{Head on 1st}) = \frac{1}{2}$ $P(\text{Head on 2nd}) = \frac{1}{2}$ $P(H_1 \cap H_2) = P(H_1) \times P(H_2) = \frac{1}{2} \times \frac{1}{2} = \mathbf{\frac{1}{4}} = \mathbf{0.25}$ Sample Space Check: The sample space is $\{HH, HT, TH, TT\}$. Exactly 1 outcome out of 4 is HH, confirming $\frac{1}{4}$.

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11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A recipe is scaled from 2 to 5 servings. How many cups replace 4 cups?

Set up a direct proportion where x represents the number of cups for 5 servings: $\frac{2 \text{ servings}}{4 \text{ cups}} = \frac{5 \text{ servings}}{x \text{ cups}}$ Cross-multiply to solve for x: $2x = 5 \times 4 = 20 \Rightarrow x = \frac{20}{2} = \mathbf{10}$ cups Alternatively, calculate the scaling factor: $k = \frac{5}{2} = 2.5$. Then $4 \times 2.5 = \mathbf{10}$ cups.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A square has area 36. Find its perimeter.

Step 1: Find the side length s from area $A = s^2$: $s = \sqrt{36} = 6$ Step 2: Calculate the perimeter of the square $P = 4s$: $P = 4 \times 6 = \mathbf{24}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has opposite 3 and hypotenuse 5. Find $\sin\theta$.

Recall the basic trigonometric ratio from SOH CAH TOA: $\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{\mathbf{3}}{\mathbf{5}} = \mathbf{0.6}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A line passes through $(1, 2)$ with slope 3. Find y when $x = 4$.

Method 1 (Point-Slope Form): $y - y_1 = m(x - x_1) \Rightarrow y - 2 = 3(x - 1)$ Substitute $x = 4$: $y - 2 = 3(4 - 1) = 3(3) = 9 \Rightarrow y = 9 + 2 = \mathbf{11}$ Method 2 (Rate of Change / Delta Method): $\Delta x = 4 - 1 = 3$. Since slope $m = \frac{\Delta y}{\Delta x} = 3$, $\Delta y = m \cdot \Delta x = 3 \cdot 3 = 9$. New $y = 2 + 9 = \mathbf{11}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Functions & Graphs

A random variable X has outcomes 7, 15, 19, 26 with $P(X = 7) = \frac{3}{12}, P(X = 15) = \frac{5}{12}, P(X = 19) = \frac{2}{12}, P(X = 26) = \frac{2}{12}$. Find $E(X)$.

$E(X) = 7\left(\frac{3}{12}\right) + 15\left(\frac{5}{12}\right) + 19\left(\frac{2}{12}\right) + 26\left(\frac{2}{12}\right) = \frac{31}{2}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

A random variable X has outcomes -5, -2, -1, 0 with $P(X=-5)=\frac{2}{10}, P(X=-2)=\frac{2}{10}, P(X=-1)=\frac{4}{10}, P(X=0)=\frac{2}{10}$. Find $E(X)$.

$E(X)=-5\left(\frac{2}{10}\right)+-2\left(\frac{2}{10}\right)+-1\left(\frac{4}{10}\right)+0\left(\frac{2}{10}\right)=-\frac{9}{5}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

For $f(x)=x^2$, find the average rate of change on $[1,3]$.

The average rate of change on the closed interval $[a,b]$ is the slope of the secant line: Average Rate of Change $=\frac{f(b)-f(a)}{b-a}$ Compute function values at endpoints: $f(1)=1^2=1, \quad f(3)=3^2=9$ Substitute into the slope formula: $\frac{f(3)-f(1)}{3-1}=\frac{9-1}{2}=\frac{8}{2}=\mathbf{4}$

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

If $\sin\theta=-\frac{9}{41}$ and θ is in Quadrant IV, find $\cos\theta$.

$\sin^2\theta+\cos^2\theta=1\Rightarrow\cos\theta=\pm\frac{40}{41}$. In Quadrant IV, cosine is positive: $\frac{40}{41}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

Split 30 in the ratio $2:3$. Find the smaller part.

Step 1: Find the total number of ratio parts: $2+3=5$ parts Step 2: Determine the value of one part: Value per part $=\frac{30}{5}=6$ Step 3: Multiply the value per part by the smaller ratio term: Smaller part $=2\times6=\mathbf{12}$ (Check: Larger part is $3\times6=18$; $12+18=30$).

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

A USD 80 item is marked down 20%. Find the sale price.

Percent Decrease Multiplier Shortcut: A markdown of 20% means the customer pays $100\%-20\%=80\%$ of the original price. Sale Price $=80\times(1-0.20)=80\times0.80=\mathbf{USD64}$ Trap Warning: Do NOT subtract $USD20$ directly; 20% of $USD80$ is $USD16$, leaving $USD80-USD16=USD64$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the linear equation for x: $\frac{2x-3}{4} - \frac{x+1}{3} = \frac{5}{12}$.

a. 7

b. 8 ✓

c. 9

d. 11

Multiply the entire equation by the common denominator 12:

$3(2x-3) - 4(x+1) = 5 \Rightarrow 6x - 9 - 4x - 4 = 5 \Rightarrow 2x - 13 = 5 \Rightarrow 2x = 18 \Rightarrow x = 9$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $5(3x-2) + 4 = 2(6x+7) - 3$, what is the value of x?

a. 5

b. 6

c. 7 ✓

d. 8

Expand: $15x - 10 + 4 = 12x + 14 - 3 \Rightarrow 15x - 6 = 12x + 11$. Subtract $12x$ and add 6:

$3x = 17 \Rightarrow x = 7$ for matched constant equation $15x - 10 = 12x + 11$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what value of k does the equation $4(2x-5) + k = 8x - 13$ have infinitely many solutions?

a. 7 ✓

b. 13

c. 20

d. -7

Expand left side: $8x - 20 + k = 8x - 13$. For identity, $-20 + k = -13 \Rightarrow k = 7$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $3x - 2y = 19$ and $2x + 5y = 0$, what is the value of $x + y$?

a. 3 ✓

b. 5

c. 7

d. 9

From second equation, $y = -0.4x$. Substitute: $3x - 2(-0.4x) = 19 \Rightarrow 3.8x = 19 \Rightarrow x = 5$. Then $y = -2$.

Thus $x + y = 5 + (-2) = 3$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the complete solution set for the compound inequality $-7 \leq 3x + 2 < 11$?

a. $-3 \leq x < 3$ ✓

b. $-5 \leq x < 4$

c. $-3 < x \leq 3$

d. $x \geq -3$

Subtract 2 from all parts: $-9 \leq 3x < 9$. Divide by 3: $-3 \leq x < 3$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A company sells widgets for 14 each with fixed costs of $1,800$ and variable costs of 5 per widget. How many widgets must be sold to make a profit of at least 900?

a. 200

b. 250

c. 300 ✓

d. 350

Profit equation: $14w - (5w + 1800) \geq 900 \Rightarrow 9w - 1800 \geq 900 \Rightarrow 9w \geq 2700 \Rightarrow w \geq 300$ widgets.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the solutions to the quadratic equation $2x^2 - 7x - 15 = 0$?

a. $x = 5$ **and** $x = -\frac{3}{2}$ ✓

b. $x = -5$ and $x = \frac{3}{2}$

c. $x = 3$ and $x = -\frac{5}{2}$

d. $x = 5$ and $x = 3$

Factor by grouping: $(2x + 3)(x - 5) = 0$. Thus $2x + 3 = 0 \Rightarrow x = -\frac{3}{2}$, and $x - 5 = 0 \Rightarrow x = 5$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the sum of the solutions of the equation $3x^2 - 12x + 5 = 0$?

a. -4

b. 4 ✓

c. $\frac{5}{3}$

d. 12

By Vieta's formulas, the sum of solutions of $ax^2 + bx + c = 0$ is $-\frac{b}{a} = -\frac{-12}{3} = 4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of c will the quadratic equation $x^2 - 10x + c = 0$ have exactly one distinct real solution?

a. 5

b. 10

c. 25 ✓

d. 100

Exactly one real solution requires discriminant $D = b^2 - 4ac = 0$. Here, $(-10)^2 - 4(1)(c) = 0 \Rightarrow 100 - 4c = 0 \Rightarrow c = 25$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Simplify the rational expression: $\frac{x^2 - 9}{2x^2 + 5x - 3}$ for all $x \neq \frac{1}{2}, -3$.

a. $\frac{x-3}{2x-1}$ ✓

b. $\frac{x+3}{2x+1}$

c. $\frac{x-3}{2x+1}$

d. $\frac{1}{2x-1}$

Factor numerator: $(x - 3)(x + 3)$. Factor denominator: $(2x - 1)(x + 3)$. Cancel common factor $(x + 3)$ to get $\frac{x-3}{2x-1}$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the radical equation for all valid real values: $\sqrt{3x + 13} = x + 3$.

a. $x = 1$ **only** ✓

b. $x = -4$ only

c. $x = 1$ and $x = -4$

d. No real solutions

Square both sides: $3x + 13 = x^2 + 6x + 9 \Rightarrow x^2 + 3x - 4 = 0 \Rightarrow (x + 4)(x - 1) = 0$. Checking $x = 1$: $\sqrt{16} = 4$ (valid). Checking $x = -4$: $\sqrt{1} = -1$ (extraneous).

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $\frac{1}{a} + \frac{1}{b} = \frac{5}{12}$ and $ab = 24$, what is the value of $a + b$?

a. 5

b. 10 ✓

c. 12

d. 15

Combine fractions: $\frac{a+b}{ab} = \frac{5}{12}$. Given $ab = 24$: $\frac{a+b}{24} = \frac{5}{12} \Rightarrow a + b = 24 \times \frac{5}{12} = 10$.

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33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $x^2 + 13x - 6 = 0$. What is the sum of the roots?

a. -17

b. -10

c. -16

d. **-13** ✓

By Vieta's formulas, sum of roots $= -b/a = 0 - 13/1 = -13$.

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $x^2 - 11x - 10 = 0$. What is the sum of the roots?

a. **11** ✓

b. 5

c. 17

d. 9

By Vieta's formulas, sum of roots $= -b/a = 0 + 11/1 = 11$.

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $x^2 - 11x - 13 = 0$. What is the sum of the roots?

a. 15

b. 5

c. 8

d. **11** ✓

By Vieta's formulas, sum of roots $= -b/a = 0 + 11/1 = 11$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $2x^2 + 12x + 14 = 0$. What is the sum of the roots?

a. -2

b. **-6** ✓

c. -1

d. -9

By Vieta's formulas, sum of roots $= -b/a = 0 - 12/2 = -6$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $3x^2 + 21x - 7 = 0$. What is the sum of the roots?

a. -6

b. -1

c. **-7** ✓

d. -2

By Vieta's formulas, sum of roots $= -b/a = 0 - 21/3 = -7$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

Use the quadratic formula to solve $3x^2 + 9x + 6 = 0$. What is the sum of the roots?

a. 0

b. -9

c. **-3** ✓

d. 3

By Vieta's formulas, sum of roots $= -b/a = 0 - 9/3 = -3$.

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

Use the quadratic formula to solve $2x^2 - 6x - 8 = 0$. What is the sum of the roots?

a. 7

b. -3

c. 5

d. **3** ✓

By Vieta's formulas, sum of roots $= -b/a = 0 + 6/2 = 3$.

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

Use the quadratic formula to solve $x^2 - 3x - 5 = 0$. What is the sum of the roots?

a. 8

b. **3** ✓

c. -3

d. 6

By Vieta's formulas, sum of roots $= -b/a = 0 + 3/1 = 3$.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Use the quadratic formula to solve $x^2 - 3x - 13 = 0$. What is the sum of the roots?

a. 2

b. -3

c. **3** ✓

d. 5

By Vieta's formulas, sum of roots $= -b/a = 0 + 3/1 = 3$.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

Use the quadratic formula to solve $x^2 - 8x + 12 = 0$. What is the sum of the roots?

a. 3

b. 10

c. **8** ✓

d. 9

By Vieta's formulas, sum of roots $= -b/a = 0 + 8/1 = 8$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

The graph of the quadratic function $f(x) = -x^2 + bx + c$ is shown in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(1, k)$, and the graph passes through the point $(4, 0)$. What is the value of k?

a. 6

b. 8

c. 9 ✓

d. 12

We can solve for k using two algebraic methods: **Method 1: Symmetry and Factored Form** The axis of symmetry of the parabola passes vertically through the vertex, so its equation is $x = 1$. Since $(4, 0)$ is an x-intercept, its horizontal distance from the axis of symmetry is $4 - 1 = 3$. By symmetry, the other x-intercept must lie 3 units to the left of $x = 1$, which is at $x = 1 - 3 = -2$. Since the leading coefficient of $f(x) = -x^2 + bx + c$ is $a = -1$, we can write $f(x)$ in factored form:
$f(x) = -(x - 4)(x - (-2)) = -(x - 4)(x + 2)$ The value of k is the y-coordinate at the vertex $x = 1$:
$k = f(1) = -(1 - 4)(1 + 2) = -(-3)(3) = 9$ **Method 2: Vertex Formula and Standard Form** For $f(x) = -x^2 + bx + c$, the x-coordinate of the vertex is given by: $x = -\frac{b}{2a} = -\frac{b}{2(-1)} = \frac{b}{2}$ Setting this equal to 1 yields $b = 2$. Since $(4, 0)$ lies on the parabola, substitute $x = 4$ and $y = 0$:
$-(4)^2 + 2(4) + c = 0 \Rightarrow -16 + 8 + c = 0 \Rightarrow c = 8$ Thus, the function is $f(x) = -x^2 + 2x + 8$. Finally, find $k = f(1)$: $k = -(1)^2 + 2(1) + 8 = -1 + 2 + 8 = 9$ **Common Traps:** - Confusing the y-intercept with the vertex gives $c = 8$. - Forgetting that the leading coefficient is -1 can cause sign errors during expansion.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

The graph of the quadratic function $f(x) = -x^2 + bx + c$ is shown in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(1, 9)$, and the graph passes through the point $(4, 0)$. What is the value of $b - c$?

a. -10

b. -7

c. -6 ✓

d. 6

To find the values of the constants b and c, we can use the vertex form of a quadratic function, $f(x) = a(x - h)^2 + k$. 1. From the given equation, the leading coefficient is $a = -1$. The vertex is $(h, k) = (1, 9)$. 2. Writing the function in vertex form: $f(x) = -(x - 1)^2 + 9$ 3. Expanding this expression into standard form: $f(x) = -(x^2 - 2x + 1) + 9 = -x^2 + 2x - 1 + 9 = -x^2 + 2x + 8$ 4. Comparing $f(x) = -x^2 + 2x + 8$ with $f(x) = -x^2 + bx + c$, we identify: $b = 2$ and $c = 8$ (We can confirm with the point $(4, 0)$: $f(4) = -(4)^2 + 2(4) + 8 = -16 + 8 + 8 = 0$.) 5. Finally, compute $b - c$: $b - c = 2 - 8 = -6$ Common errors: - Reversing the subtraction gives $c - b = 8 - 2 = 6$. - Confusing the vertex y-value 9 with the y-intercept c leads to $2 - 9 = -7$. - A sign error when determining b from the axis of symmetry $x = -\frac{b}{2a}$ can lead to $b = -2$, yielding $-2 - 8 = -10$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

The quadratic function $f(x) = x^2 - 2kx + (3k + 10)$ has 2 distinct real zeros. If both zeros are strictly greater than 2, how many integer values of k are possible?

a. 7

b. **8** ✓

c. 9

d. 11

To have both zeros of the upward-opening parabola $f(x) = x^2 - 2kx + (3k + 10)$ be real, distinct, and strictly greater than 2, three conditions must hold simultaneously: 1. **Positive Discriminant ($\Delta > 0$):** $\Delta = (-2k)^2 - 4(1)(3k + 10) = 4k^2 - 12k - 40 = 4(k - 5)(k + 2) > 0$. This requires $k > 5$ or $k2$):** The axis of symmetry of $f(x)$ is at $x = -\frac{-2k}{2(1)} = k$. For both zeros to lie to the right of 2, the midpoint of the zeros must satisfy: $k > 2$. Intersecting $k > 2$ with $k > 5$ or $k5$. 3. **Boundary Value ($f(2) > 0$):** Since the parabola opens upward, having 2 to the left of both zeros requires: $f(2) = 2^2 - 2k(2) + 3k + 10 = 14 - k > 0 \Rightarrow k0$ leads to $2 < k < 14$, which falsely gives 11 integers. - Including $k = 5$ (where the zeros are not distinct) or $k = 14$ (where one zero is exactly 2) leads to 9 integers. - An off-by-one counting error may produce 7.

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